

Workflow and architecture patterns for working together 2: Architectures

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Outline

1 Current challenges and their canonical solutions

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 - Theory
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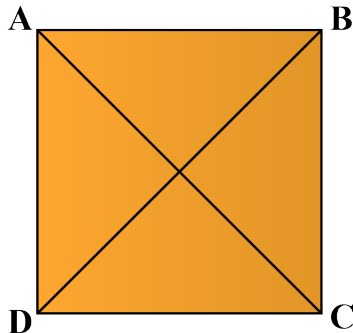
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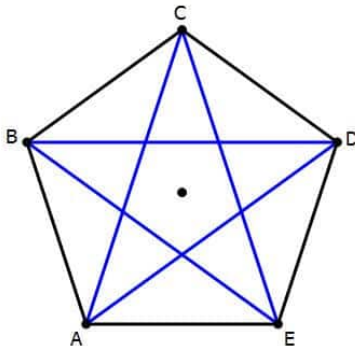
1975: The Mythical Man Month, Fred Brooks

- number of direct communication paths for n individuals= $n(n-1)/2$



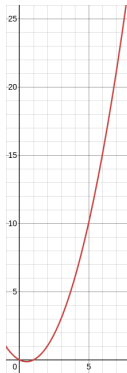
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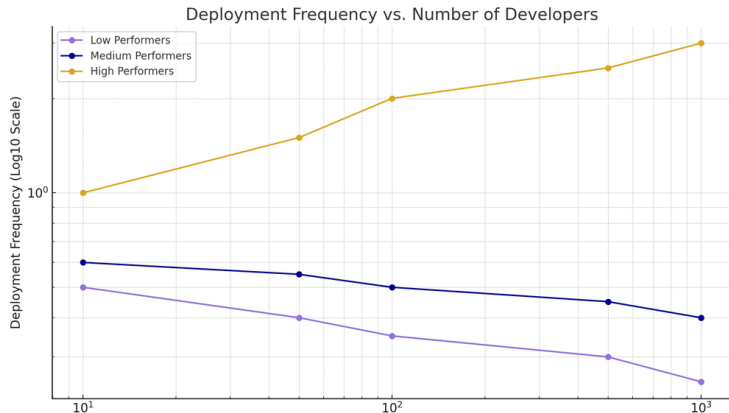
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$$n(n-1)/2$$



The Mythical Man Month, Fred Brooks (cont.)

- corollary: adding more people to a project can lead not only to diminishing returns on delivery speed, but to objectively less work being completed



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1967: Conway's law

'Organizations which design systems [...] are constrained to produce designs which are copies of the communication structures of these organizations.' - Melvin Conway, 'How do Committees Invent?' *Datamation*, 1967

Conway's law: examples

- separate .csproj projects for tests written by QA
- separate development and config repositories for the same transaction signifying low communication between development and operations
- all business documentation in the same repository, separate from the repositories that the documentation is for

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- Teams working in the same part of the same codebase at the same time are the same team
 - separate the codebases for improved velocity
 - join the teams for improved code sharing and communication (Inverse conway maneuver)
 - branching-based strategies for team management compromise between the above approaches without their benefits

Revisiting Conway's law: defining a team

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A naive probability theory for merge consistency: isolated commits

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A naive probability theory for merge consistency: modeling branches as sets of commits

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 - $= \frac{1}{2^n - 2^{n_1} - 2^{n_2} - 2^{n_3} + 3}$

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A naive probability theory for merge consistency: general definition for distinct branches diverging from a common ancestor

- Let A be the set of unmerged commits across k distinct branches $A_1 \cdots A_k$, where $A = A_1 \sqcup A_2 \sqcup \cdots \sqcup A_k$, $n = |A|$, and $n_i = |A_i|$ for $1 \leq i \leq k$. Then

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$$P(\text{no conflicts}) = \frac{1}{2^n - (\sum_{i=1}^k 2^{n_i}) + k}$$

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The State of DevOps Report: throughput metrics

- lead time for changes

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- deployment frequency

The State of DevOps Report: stability metrics

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The State of DevOps Report: results

'Astonishingly, these results demonstrate that there is no trade-off between improving performance and achieving higher levels of stability and quality. Rather, high performers do better at all of these measures. This is precisely what the Agile and Lean movements predict, but much dogma in our industry still rests on the false assumption that moving faster means trading off against other performance goals, rather than enabling and reinforcing them.'
- Nicole Forsgren, Jez Humble, and Gene Kim, Accelerate (2017), p. 14

The State of DevOps Report: results

When compared to low performers, elite performers realize

127x

faster lead
time

182x

more
deployments
per year

8x

lower change
failure rate

2293x

faster failed
deployment
recovery times

- 2024 State of DevOps Report

Approximating DevOps metrics in our current environment

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Throughput: overall velocity metrics

Branch	Runs	Earliest run	Contributors
NG_Claims_837D_COB	74	23 Mar	5
PEDI-35120_837D_COB	85	15 Apr	5
Either COB branch	159	23 Mar	6
All other branches	123	23 Mar	12
	72	15 Apr	10
	266	21 Jan	

Source: GH Actions Simulation tests, 28 May 2026

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Stability: pipeline failure rate metrics

Branch	Simulation test failures	Unit test failures	Earliest commit
NG_Claims_837D_COB	56/74 (76%)	56/74 (76%)	23 Mar
PEDI-35120_837D_COB	20/85 (23.5%)	21/85 (24.7%)	15 Apr
Either COB branch	76/159 (48%)	77/159 (48%)	23 Mar
All extant branches	73/123 (59%)	87/123 (71%)	23 Mar
	50/72 (69%)	51/72 (71%)	15 Apr
	92/161 (57%)	104/161 (64%)	21 Jan

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Source: GH Actions, 28 May 2026

- 28% better simulation test pass rate since 23 Mar

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PEDI-35120_837D_COB	20/85 (23.5%)	21/85 (24.7%)	15 Apr
Either COB branch	76/159 (48%)	77/159 (48%)	23 Mar
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- Source command:

```
git log --since="2026-04-15"  
--oneline <--merges> <--all>  
<^PEDI-35120_837D_COB> | <grep -iv 'merge pull  
request'> | wc -l
```


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- pipeline is definitive for deployment - daily stale branch job

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